Seokyung **Park**

Agricultural Sensing and Plant Phenotyping

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Research interests

My research interests focus on image-based analysis and sensing for agricultural applications, particularly plant phenotyping. I aim to integrate imaging, environmental sensing and robotic platforms to quantify crop traits and understand their relationships with growing conditions.

Education

Pusan National University Mar. 2025 – Feb. 2027 (expected)

M.S. in Bio-Industrial Machinery Engineering | GPA: 4.36/4.50

Pusan National University Mar. 2017 – Feb. 2025

B.S. in Plant Bioscience | GPA: 4.26/4.50

Minor in Bio-Industrial Machinery Engineering

Summa Cum Laude; President's List

Research experience

Precision Agriculture and Smart Sensing System Lab., Pusan National University

Jun. 2024 – Present

Roles: Graduate Researcher, Mar. 2025 – Present; Undergraduate Research Assistant, Jun. 2024 – Feb. 2025

Damage-aware robotic gripping for tomato harvesting

- Developed Python software for robotic compression experiments and force-signal processing using a myCobot 320 Pi, an adaptive gripper and a force sensor; evaluated four loading rates ranging from 6.9 to 15.3 N/s.
- Combined force-deformation measurements with hyperspectral imaging at 1 hour and 5 days after compression to assess immediate and delayed tomato damage.
- Contributed to experimental design, data analysis and manuscript preparation as first author. The resulting manuscript is under review at *Biosystems Engineering*.

Quadruped robotics for agricultural sensing

- Conducted greenhouse microclimate mapping using a Unitree Go2 quadruped robot, measuring temperature, relative humidity and CO_2 at two canopy heights across eight crop lanes.
- Produced 24 spatial maps to evaluate the representativeness of fixed greenhouse sensors and contributed to the resulting manuscript as a co-first author.

Quadruped-based 3D plant phenotyping

- As my M.S. thesis research, I am developing a mobile RGB-D workflow for close-range 3D plant reconstruction using a RealSense D456 mounted on a Unitree Go2 quadruped robot.
- Implemented depth filtering, IMU- and odometry-based pose correction, point-cloud registration, and trajectory optimization; strengthened my ability to assess reconstruction quality by evaluating local consistency using vegetation overlapping and nearest-neighbor distance.

Laboratory project: Development of an Integrated Sensing, Monitoring, and Decision-Making Model for Field Surveys in Agricultural Environments Using a Quadruped Robot (NRF/MSIT; participant).

Quadruped-based upstream reservoir surveys

- Conducted close-range upstream soil surveys using a quadruped robot, gaining field experience in robotic data collection outside controlled greenhouse settings.
- Combined RGB imaging and RTK-GNSS data to produce location-linked ground-surface datasets, developing practical skills in integrating visual and positioning data for field surveys.

Contract research: Quadruped-based Close-range Upstream Soil Survey under the Integrated Technology Development for Reservoir Embankment Risk Prediction and Maintenance Management Project, Phase I (NDMI; completed).

Air-dome foundations and structural sensing

- Uplift resistance of expandable foundation piles: Modelled wing inclination and dimension effects in Abaqus/Standard and analyzed uplift load–displacement behavior and soil plastic zones in sandy soil, strengthening my ability to interpret soil–structure interaction through finite-element analysis.
- Load-cell sensing for air-supported domes: Installed load cells and acquired cable-tension data at Damyang (15 June 2026) using an indicator, USB-to-RS485 interface, and Raspberry Pi. This work strengthened my practical skills in assembling and operating a field measurement system.

Laboratory project: Development of a Commercial-scale, Low-energy Vertical Farming System Based on Passive Air-dome Technology for Hot Climates in the Middle East (IPET/MAFRA; participant).

Functional Crop Laboratory, Pusan National University

Jan. 2023 – Mar. 2024

Role: Undergraduate Research Student

- Investigated soybean leaf phenotyping under drought using time-series RGB images, YOLOv8, and image-processing methods, with environmental measurements collected using Arduino-based sensors; developed foundational skills in image-based plant phenotyping and environmental sensing.

Publications and manuscripts

Peer-reviewed publication

[1] J. Park, **S. Park**, E. Jung, J. Song, G. Park, J. Lee, M. Kim, and J. Park. Inertial sensor applications in agriculture: Sensor types and data utilization techniques for mechanical and wearable applications. *Korean Journal of Agricultural Science*, 52(4):447–469, 2025. doi: 10.7744/kjoas.520405.

Under review

Park, S., Song, J., Hwang, H., Park, G., and Park, J. (2026). Speed-dependent control of robotic gripper for tomato harvesting to minimize compression-induced damage. *Biosystems Engineering*. **Under review.**

Park, S., Park, J., Jang, J., Park, D., Kwon, S., Kim, M., and Choi, G. (2026). Analysis of uplift resistance characteristics of expandable piles in sandy soil with varying wing inclination and dimensions. *Journal of the Korean Society of Agricultural Engineers*. **Under review.**

Manuscripts in preparation

Hwang, H.[†], **Park, S.**[†], and Park, J. Greenhouse microclimate mapping using a quadruped robot to assess fixed sensor representativeness. [†]Equal contribution.

Research funding

- **Master's Student Research Grant:** National Research Foundation of Korea (NRF), funded by the Ministry of Education (MOE) Grant RS-2025-25438074
  Selected for a one-year individual research award under the NRF Research Subsidies for Master's Students program.
  Title: Development of Locomotion Algorithms and Autonomous Control Systems for Quadruped Robots Specialized for Korean Agricultural Terrain.

Technical skills

- **Programming and data analysis:** Python (NumPy, pandas and SciPy) and C/C++; time-series processing, nonlinear model fitting, statistical analysis and research visualization.
- **Robotics and mobile sensing:** ROS 2, Unitree Go2, RTK-GNSS, LiDAR and IMU; robot–sensor integration, time-synchronized data acquisition and location-linked data processing.
- **Instrumentation and data acquisition:** Force sensors, load cells, Raspberry Pi, Arduino and RS-485; experimental control, sensor calibration and CSV-based data logging.
- **Crop imaging and analysis:** RGB and hyperspectral imaging; Segment Anything Model (SAM) and YOLOv8, image segmentation, spectral comparison and time series crop-image analysis.
- **3D reconstruction for plant phenotyping:** RGB-D data acquisition using a RealSense D456; depth processing, point cloud generation and registration, pose alignment using IMU and robot odometry, and evaluation of registration consistency.
- **Finite-element modelling:** Abaqus/CAE and Abaqus/Standard; contact modeling between piles and soil, simulations under displacement control, parameter studies and interpretation of soil plastic zones.

Selected conference presentations

- **Park, S.** (14 July 2026). Height-resolved 3D microclimate mapping and spatiotemporal analysis within greenhouse crop canopies using a quadruped robot. *AKABFE Student Research Spotlight, ASABE Annual International Meeting*, Indianapolis, USA. Presenter.
- **Park, S.**, Park, J., Hwang, H., and Kang, S. (2026). Ground-reaction-force-based lateral slope estimation for furrow-following quadruped robots. *Korean Society for Agricultural Machinery Spring Conference*.
- **Park, S.**, Park, J., Song, J., and Kim, M. (31 October 2025). Analysis of pile pull-out behavior in Middle Eastern soil using digital image correlation. *Korean Society of Agricultural Engineers Autumn Conference*, Jeju.
- Hwang, H., **Park, S.**, et al. (22 May 2025). Remote control and integrated sensor-data environment for an agricultural quadruped robot. *Korean Society for Precision Agriculture Spring Conference*. Co-author.
- **Park, S.**, Park, J., Park, G., Song, J., and Hwang, H. (25 April 2025). Minimizing damage and optimizing control in fruit and vegetable harvesting using a robotic arm and gripper with real-time force sensing. *Korean Society for Agricultural Machinery Spring Conference*.

Honours and awards

- **Best Paper Presentation Award (team)**, Korean Society for Precision Agriculture, May 2025 and November 2024.
- **Excellent Paper Presentation Award**, Korean Society for Agricultural Machinery, April 2025 and October 2024.
- **Excellence Award**, PNU–PKNU Field Problem Research Group Final Symposium, November 2024.
- **Best Presentation Award (team)**, Korean Society for Engineering Education, September 2024.
- **Bronze Prize**, PNU College of Natural Resources and Life Science Capstone Design Competition, October 2023.
- **Excellence Award**, Busan LINC 3.0 Entrepreneurship Shared University Competition, January 2023.
- **PNU merit scholarships**: four full-tuition and two partial-tuition awards, 2017–2024.

Selected coursework

Graduate: Biological Image Processing; Statistical Processing and Data Analysis in Bio-Industrial Machinery Engineering; Advanced Bio-Industrial Machinery Design; Biosystems Mechanics; Advanced Biomaterials Engineering I.

Undergraduate – engineering and sensing: Biosensor and Bio-Instrumentation & Practice; Biosystems Control Engineering; Bio-Production Mechatronics Design; Applied Mechanics of Materials; Bio-Industrial Machine Element Design (I); Food Process Engineering & Practice.

Undergraduate – programming and quantitative methods: Basic Computer Programming; Experimental Design for Biologists; Biostatistics; Calculus (II).

Undergraduate – plant science and cultivation: Plant Physiology and Practice; Plant Genetics; Plant Molecular Biology; Plant Biotechnology; Plant Breeding (I, II); Soil Science; Plant Ecology; Protected Horticulture.

Additional training

Modeling Contact with Abaqus/Standard, VPK PLM Solution Training Center, 25–26 June 2026.
Introduction to Abaqus/CAE, VPK PLM Solution Training Center, 18–20 March 2026.